

COURSE SYLLABUS

BOROUGH OF MANHATTAN COMMUNITY COLLEGE

The City University of New York

Department of Computer Information Systems

Title of Course: Data Structures

Class Hours per Week: 3

CSC 331

Lab Hours per Week: 2

Fall 2025

Credits: 4

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Course Description: This course will introduce students to linear and non-linear data structures, their use and implementation, algorithms, and software engineering techniques. Topics will include: stacks, queues, linked lists, hash tables, trees, graphs, searching and sorting techniques. Asymptotic analysis of algorithms and data structures will also be discussed.

Basic skills:

ENG 095, ESL 095, ACR 095.

Prerequisites:

CSC 211 (Advanced Programming Techniques) and CSC 231 (Discrete Structures) or Departmental approval.

Student Learning Outcomes:

Course Student Learning Outcomes (Students will be able to...)	Measurements (means of assessment for student learning outcomes listed in first column)
1. Demonstrate knowledge of various data structures and implement these structures in a high level language.	1. Short essay questions in exams, programming questions in exams and programming projects.
2. Use recursion for the implementation of algorithms.	2. Programming projects and exam questions.
3. Analyze and compare the performance of various sorting algorithms in terms of “Big-O” complexity and space requirements and be able to implement the algorithms.	3. Exam questions and programming projects.

5. Analyze a problem and determine the appropriate data structure for the solution of the problem.	5. Exams questions and programming projects.
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General Education Outcomes and Assessment:

	General Education Learning Outcomes	Measurements (means of assessment for general education goals listed in first column)
X	Communication Skills- Students will be able to write, read, listen and speak critically and effectively.	Provide documentation for all programming projects and short essay questions.
X	Quantitative Reasoning- Students will be able to use quantitative skills and the concepts and methods of mathematics to solve problems.	Use formulas and concepts of mathematics to solve problems in programming assignments and exams.
	Scientific Reasoning- Students will be able to apply the concepts and methods of the natural sciences.	
	Social and Behavioral Sciences- Students will be able to apply the concepts and methods of the social sciences.	
	Arts & Humanities- Students will be able to develop knowledge and understanding of the arts and literature through critiques of works of art, music, theatre or literature.	
X	Information & Technology Literacy- Students will be able to collect, evaluate and interpret information and effectively use information technologies.	Use a high-level computer programming language to create application software.
	Values- Students will be able to make informed choices based on an understanding of personal values, human diversity, multicultural awareness and social responsibility.	

Required Text & Readings

Textbook **Data Structures using C++, Second Edition**
Author: **D.S. Malik**
Publisher: **Course Technology**
ISBN-13: **978-0-324-78201-1**
ISBN-10: **0-324-78201-1**

Other Resources

Flash drives are recommended for backups of your course work.

Evaluation & Requirements of Students:

Exam 1	25%
Exam 2	25%
Final Exam	25%
Homework/Projects	<u>25%</u>
Total	100%

Outline of Topics

Week No.	Topic
Week 1	Overview/Review
Week 2	Array-Based Lists
Week 3	Linked lists I
Week 4	Linked Lists II
Week 5	Recursion EXAM1
Week 6	Stacks and Queues
Week 7	Queues
Week 8	Searching and Hashing
Week 9	Sorting Algorithms
Week 10	Sorting Algorithms EXAM2
Week 11	Binary Trees
Week 12	Binary Search Trees and AVL Trees
Week 13	Graphs I
Week 14	Graphs II and Review
Week 15	Final Exam

Class Participation

Participation in the academic activity of each course is a significant component of the learning process and plays a major role in determining overall student academic achievement. Academic activities may include, but are not limited to, attending class, submitting assignments, engaging in in-class or online activities, taking exams, and/or participating in group work. Each instructor has the right to establish their own class participation policy, and it is each student's responsibility to be familiar with and follow the participation policies for each course.

BMCC is committed to the health and well-being of all students. It is common for everyone to seek assistance at some point in their life, and there are free and confidential services on campus that can help.

Single Stop www.bmcc.cuny.edu/singlestop, room S230, 212-220-8195, singlestop@bmcc.cuny.edu. If you are having problems with food or housing insecurity, finances, health insurance or anything else that might get in the way of your studies at BMCC, come by the Single Stop Office for advice and assistance. Assistance is also available through the Office of Student Affairs, S350, 212-220-8130, studentaffairs@bmcc.cuny.edu.

Counseling Center www.bmcc.cuny.edu/counseling, room S343, 212-220-8140, counselingcenter@bmcc.cuny.edu. Counselors assist students in addressing psychological and adjustment issues (i.e., depression, anxiety, and relationships) and can help with stress, time management and more. Counselors are available for walk-in visits.

Office of Compliance and Diversity <https://www.bmcc.cuny.edu/about-bmcc/compliance-diversity>, room S701, 212-220-1236. BMCC is committed to promoting a diverse and inclusive learning environment free of unlawful discrimination/harassment, including sexual harassment, where all students are treated fairly. For information about BMCC's policies and resources, or to request additional assistance in this area, please visit or call the office, or email olevy@bmcc.cuny.edu, or twade@bmcc.cuny.edu. If you need immediate assistance, please contact BMCC Public safety at 212-220-8080.

Office of Accessibility www.bmcc.cuny.edu/accessibility, room N360 (accessible entrance: 77 Harrison Street), 212-220-8180, accessibility@bmcc.cuny.edu. This office collaborates with students who have documented disabilities, to coordinate

support services, reasonable accommodations, and programs that enable equal access to education and college life. To request an accommodation due to a documented disability, please visit, call the office or email.

Gifts. Receiving a gift from a student or family member may cause a conflict of interest. Receiving a gift may imply that special consideration will be given to a student. For this reason, the professor will not accept a gift from a student or family member.

BMCC Policy on Plagiarism and Academic Integrity Statement

Plagiarism is the presentation of someone else's ideas, words or artistic, scientific, or technical work as one's own creation. Using the idea or work of another is permissible only when the original author is identified. Paraphrasing and summarizing, as well as direct quotations, require citations to the original source. Plagiarism may be intentional or unintentional. Lack of dishonest intent does not necessarily absolve a student of responsibility for plagiarism. Students who are unsure how and when to provide documentation are advised to consult with their instructors. The library has guides designed to help students to appropriately identify a cited work. The full policy can be found on BMCC's Web site, www.bmcc.cuny.edu. For further information on integrity and behavior, please consult the college bulletin (also available online).

Camera Use Guidance for Online and Hybrid Courses

Faculty may require students to turn on their cameras in online and hybrid courses during class periods or for remote testing purposes.

Requiring students to use cameras may be important pedagogically to increase engagement, social connection, accountability, and collaboration. Moreover, in some testing situations, cameras are essential to ensuring the integrity of the testing environment.

CUNY is committed to using digital technology in ways that are sensitive to issues of student privacy. To ensure that students are aware that camera use may be required, a notice will be provided in CUNYfirst for all online and hybrid classes. Faculty who intend to require camera use should also clearly state this on course syllabi and cover the requirement with students on the first day of class.

There may be occasions when an otherwise in-person class is scheduled to be delivered remotely on a limited basis because of a temporary University or College pivot to remote instruction or because of faculty need. In these circumstances, faculty may encourage but not require students to turn on their cameras for purposes other than testing unless required camera use is clearly stated on the course syllabus and covered with the students on the first day of class.

Per CUNY's [verification of enrollment policy](#), students can meet the course engagement threshold for purposes of Title IV financial aid by participating in an online discussion about an academic matter, engaging in an online academically-related activity, or initiating contact with the instructor to ask a question about the academic subject studied in the course or course-related question. None of these criteria requires the use of a live camera.

Issued by

Wendy F. Hensel

Executive Vice Chancellor and University Provost

June 20, 2024